AEGIS CYBER-FORENSICS LABORATORY

DIGITAL MEDIA AUTHENTICATION & DEEPFAKE INVESTIGATION DOSSIER

Case Reference:	AEGIS-DF-00001	**Date of Analysis:**	2026-09-22 20:40:27
Investigator/User:	System Administrator	**Media Type:**	IMAGE
Original Filename:	fake.png	**Face Detected:**	Yes (Primary ROI isolated)
SHA-256 Hash:	N/A		

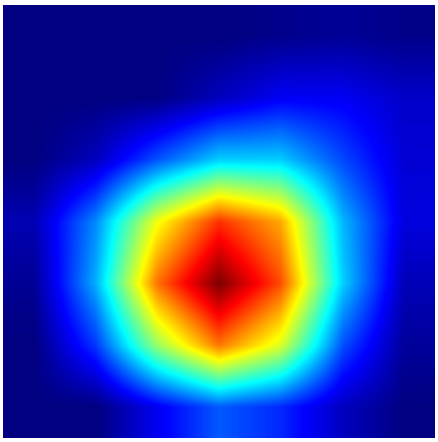
FORENSIC VERDICT: DEEPFAKE DETECTED

Confidence Level: **98.8%** | Fake Probability: **98.8%** | Real Probability: **1.2%**

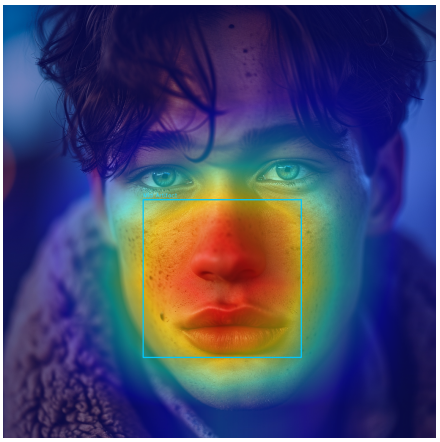
Visual Evidence & Explainable AI (Grad-CAM) Activation Maps



Isolated Facial ROI



Grad-CAM Heatmap



Suspicious Region Overlay

Forensic Anomaly & Artifact Distribution Breakdown

Forensic Indicator	Score / Value	Benchmark / Anomaly Criteria
Boundary Artifact Discontinuity	36.2%	> 35% indicates blending seams
High-Frequency Texture Variance	32.5%	Disparity from natural sensor noise profile
Facial Bilateral Symmetry Coherence	66.8%	Structural landmark alignment discrepancy
Risk Index Classification	**CRITICAL**	Computed aggregate deepfake probability

Investigative Notes & Legal Advisory

This examination was performed using a calibrated deep convolutional neural network (ResNet-18 architecture) trained on manipulated facial sequences and validated against synthetic artifact distributions. The class activation maps highlight pixel regions contributing predominantly to the model classification. This certificate provides probabilistic evidentiary support for cyber-investigations and dispute resolution.

Status: Digitally Certified

Verification Stamp: AEGIS-LABS-OK